

Dazodalibep in Sjögren's disease: forecasting the Phase 3 readout, and what could move it

Clinical deep dive on Amgen's Phase 3 program

- Primary forecast: NCT06104124 — systemic disease activity
- Supporting assessment: NCT06245408 — symptom burden
- Analysis cutoff: September 15, 2026
- Prepared by: Sai Adarsh Kasula, Data Scientist / Biostatistician

We expect NCT06104124 to meet its ESSDAI primary endpoint, with meaningful residual risk

- 1. **Direct clinical evidence:** Phase 2 showed a -2.2-point adjusted ESSDAI difference at day 169; p=0.0167.
- 2. **Biological coherence:** CD40L blockade reduced immune biomarkers (CXCL13, RF); iscalimab provides additional pathway evidence.
- 3. **Confirmatory precedent:** ianalumab demonstrated that week-48 ESSDAI benefit can be detected in Phase 3.

MAIN RISK

A smaller week-48 effect combined with higher variance or stricter multiplicity testing could turn an encouraging signal into a negative primary result.

Forecast	Conclusion	Basis / boundary
Systemic trial	Expect positive	Qualitative analyst forecast; stress-tested below
Symptomatic trial	Positive ESSPRI lean	DASPRI and full success-rule uncertainty limit conviction

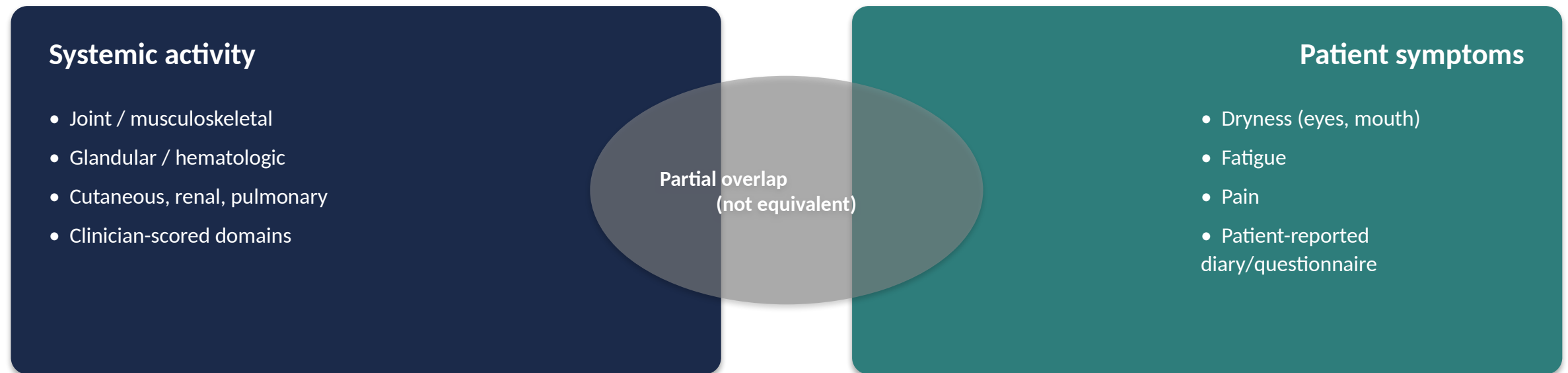
The program tests two distinct clinical questions at week 48

	Systemic: NCT06104124	Symptomatic: NCT06245408
Clinical focus	Moderate-to-severe systemic activity	High symptoms with limited systemic activity
Core eligibility	ESSDAI ≥ 5	ESSPRI ≥ 5 ; ESSDAI < 5
Enrollment	651	434
Primary outcomes	ESSDAI change from baseline	ESSPRI AND DASPRI change from baseline
Primary assessment	Week 48	Week 48
Status	Completed, primary completion 2026-07-23	Active, not recruiting; primary completion est. 2026-10-22
This presentation	Full efficacy forecast	Directional supporting assessment

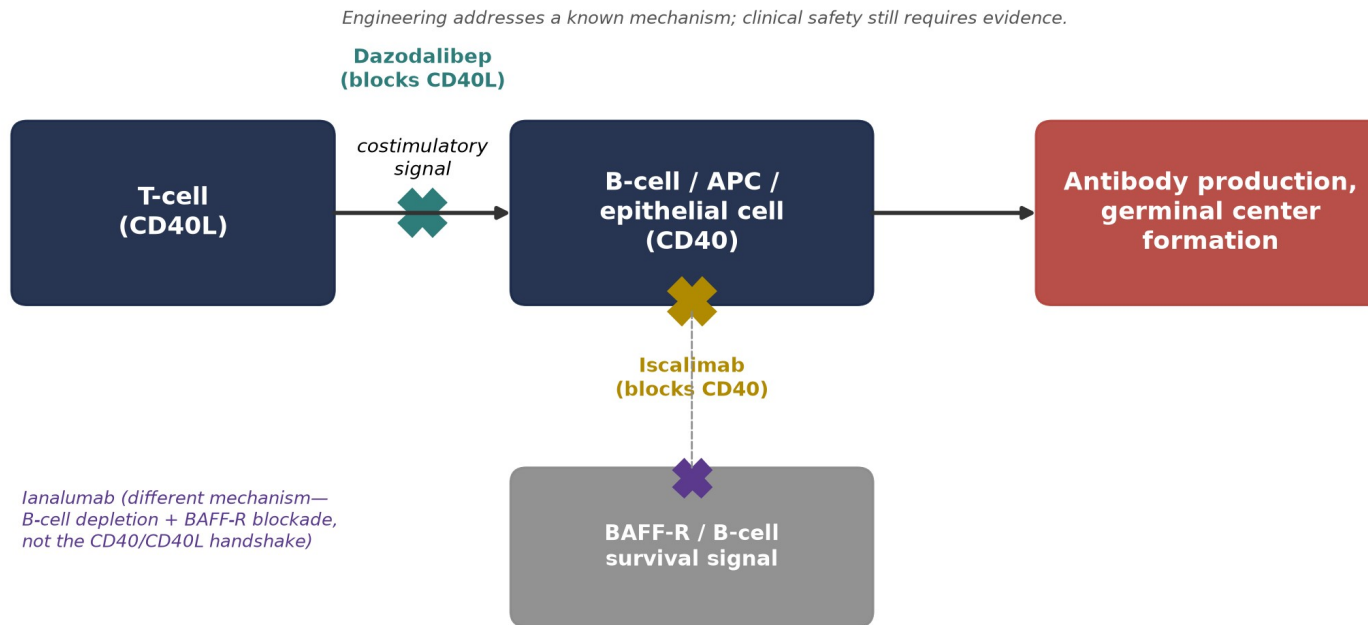
“Positive” means success under the prespecified primary testing procedure. Statistical significance, clinical importance, and benefit-risk are assessed separately.

Reducing systemic activity and improving dryness, fatigue and pain are related — but separate — tests

- **ESSDAI:** clinician-assessed systemic disease activity across weighted organ domains.
- **ESSPRI/DASPRI:** patient-reported dryness, fatigue, and pain.
- Symptoms can reflect several processes; immune suppression need not produce the same benefit in every domain.

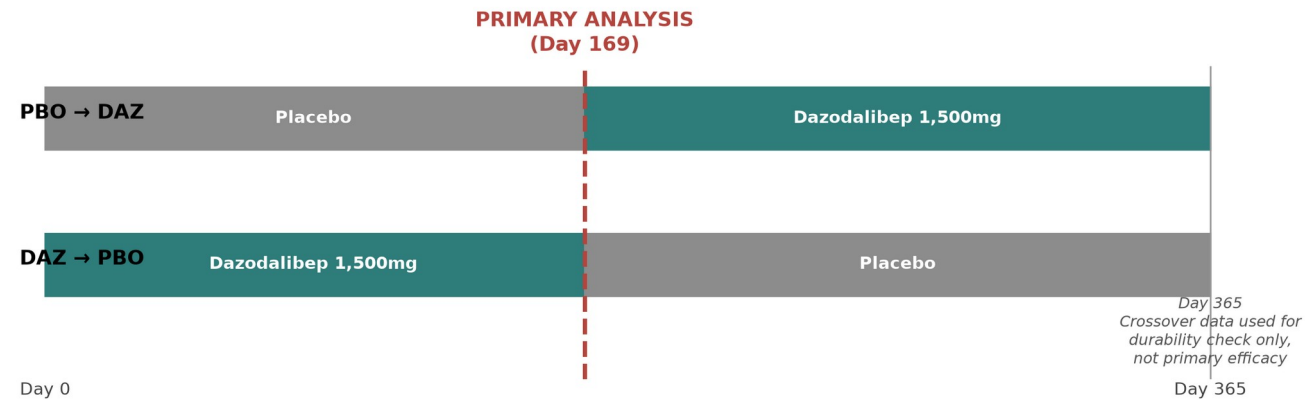


CD40L blockade interrupts immune-cell costimulation implicated in Sjögren's disease



- **CD40L-CD40 signaling** supports T-cell/B-cell interactions and downstream immune activation.
- **Dazodalibep blocks the ligand (CD40L)**; iscalimab targets the receptor (CD40).
- **Fc-free Tn3-scaffold design** is engineered to avoid the Fc-mediated platelet-crosslinking mechanism implicated in first-generation anti-CD40L antibody thromboembolism.

Phase 2 provides two randomized efficacy tests, followed by a blinded treatment switch



	Population 1	Population 2
Randomized N	74	109
Dazodalibep / placebo	36 / 38	54 / 55
Core entry criteria	ESSDAI ≥ 5	ESSPRI ≥ 5 , ESSDAI < 5 , stimulated flow ≥ 0.1 mL/min
Primary outcome	ESSDAI change, day 169	ESSPRI change, day 169
Stage I completion	71/74	102/109

One IV regimen — 1,500mg every 2 weeks for 3 doses, then every 4 weeks for 4 doses. No dose-ranging in this Sjögren's study.

Dazodalibep produced greater ESSDAI improvement than placebo in Phase 2

Day-169 change	Dazodalibep	Placebo
Adjusted mean $\pm$ SE	-6.3 $\pm$ 0.6	-4.1 $\pm$ 0.6

ADJUSTED DIFFERENCE: -2.2 POINTS

Reported 90% CI approximately -3.6 to -0.7; $p=0.0167$. This is a 90% CI — not the 95% CI used elsewhere in this deck.

- The prespecified primary result supports systemic efficacy.
- The trial used two-sided $\alpha=0.10$ (small-N caution); this result also clears standard 0.05.
- Small sample size leaves uncertainty about the reproducible effect.

READING THE EVIDENCE

- Higher ESSDAI = worse disease; a larger negative change = more improvement.
- Both arms improved from baseline — the placebo arm's own -4.1-point drop reflects the known placebo-response pattern in this endpoint.
- The 90% CI excludes zero, consistent with the reported $p=0.0167$.
- N-at-risk and the full ESSDAI trajectory (paper Fig. 2a) are not reproduced numerically here beyond the day-169 readout above.

Responder and subgroup results support a cautious interpretation of the systemic signal

EXHIBIT A — DAY-169 RESPONSE

Improvement threshold	DAZ	Placebo	p-value
≥3 ESSDAI points	26/36; 72.2%	22/37; 59.5%	0.3283
≥4 ESSDAI points	24/36; 66.7%	18/37; 48.6%	0.1823
≥5 ESSDAI points	61.1%	35.1%	0.0449; post hoc

ON-SLIDE TAKEAWAY

Directionally favorable results do not establish a robust responder effect or a treatment-by-subgroup interaction.

Bars show n/N, not just %, to keep small-sample denominators visible — e.g. 26/36 (72.2%) vs. 22/37 (59.5%) at the ≥3-point threshold. None of the three dichotomized thresholds reaches significance at the prespecified level; only the post hoc ≥5-point cut does.

EXHIBIT B

Exploratory adjusted difference without RA/SLE ≈−1.6; with RA/SLE ≈−3.3. Small subgroup Ns.

exploratory, small N

The symptom cohort showed a broad signal across dryness, fatigue and pain

EXHIBIT — POPULATION 2, DAY 169

Outcome	DAZ change	Placebo change	p-value
ESSPRI total — primary	-1.80	-0.53	0.0002
Dryness	-1.9	-0.8	0.0066
Fatigue	-1.7	-0.3	0.0022
Pain	-1.8	-0.4	0.0010

PRIMARY ENDPOINT CONTRAST

Adjusted total-score contrast -1.27; 90% CI -1.82 to -0.73. This is a 90% CI, consistent with Slide 7.

READING THE EVIDENCE

- All four rows favor dazodalibep, and the primary ESSPRI-total result is the strongest p-value in the whole Phase 2 program.
- Directional consistency across dryness, fatigue and pain supports read-through beyond the single total score.
- Domain-level tests (dryness/fatigue/pain) are supportive and unadjusted for multiplicity — read them as consistency checks, not independently confirmatory results.
- Phase 3 tests a later timepoint (week 48, not day 169) and adds DASPRI as a second co-primary (see Slide 18).

Biomarker suppression supports drug activity; crossover patterns add qualified evidence

EXHIBIT A — ADJUSTED GEOMETRIC MEAN RATIO TO BASELINE, DAY 169

Biomarker	Pop 1: DAZ/Pbo	Pop 2: DAZ/Pbo
CXCL13	0.67 / 1.00	0.56 / 1.06
Rheumatoid factor	0.66 / 1.13	0.54 / 0.94

Ratios <1.0 = suppression relative to each arm’s own baseline (not a direct drug/placebo ratio).

EXHIBIT B — CHANGE FROM BASELINE, DAY 169 → DAY 365 (CROSSOVER)

Population / sequence	Trajectory
Systemic, placebo→DAZ	ESSDAI -4.1 → -6.3
Systemic, DAZ→placebo	ESSDAI -6.3 → -4.4
Symptomatic, DAZ→placebo	ESSPRI -1.8 → -1.9

ON-SLIDE TAKEAWAY

Pharmacological activity is supported; durable clinical efficacy under the Phase 3 regimen remains a separate question.

- Switching to dazodalibep (placebo→DAZ) improved further; switching off it (DAZ→placebo) partially reversed — both consistent with an active drug effect.
- A crossover after unblinding is not a 48-week parallel-group replication; carryover and expectation effects are real limitations.

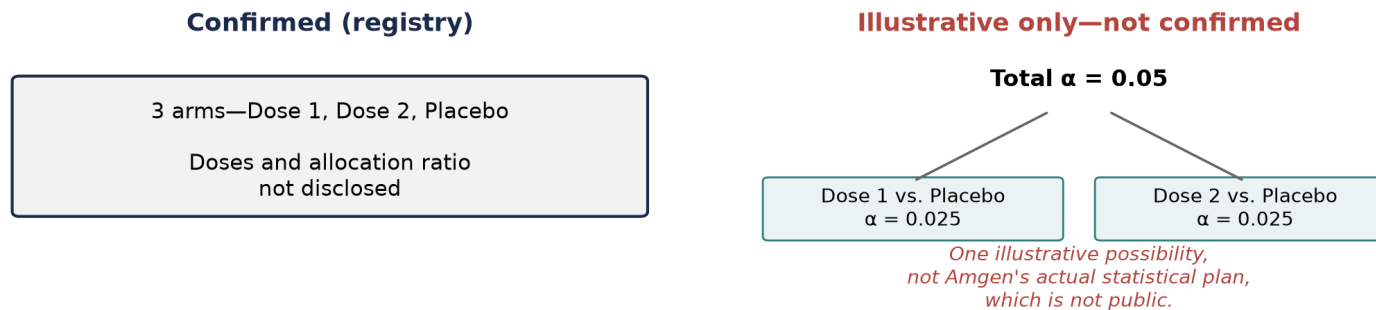
Phase 3 increases information and changes duration and dose comparisons

Factor	Phase 2 systemic	Phase 3 systemic	Forecast implication
Randomized total	74	651	More precision; not proportional effect growth
Active/control n (one comparison)	36 / 38	~217 / 217 assumed	≈6.0× per-arm increase
Primary assessment	Day 169	Week 48	Longer durability and placebo-response window
Active regimens	One dose	Two dose arms	Dose selection and multiplicity matter
ESSDAI entry floor	≥5	≥5	No established loosening of this threshold
Baseline/domain mix	Observed	Not established	A real transportability uncertainty

 = assumed or not established, not confirmed by the registry

PK/PD supports dose selection, while clinical effect and testing rules remain separate

- A population PK/PD model pooled 4 trials (healthy volunteers, RA, Sjögren's), relating exposure to immune biomarkers.
- Sjögren's patients clear the drug more slowly than healthy volunteers/RA patients — a real, population-specific PK finding.
- The model supports the proposed Phase 3 doses achieving sustained exposure above the Ki67+/RF biomarker IC50 — a pharmacodynamic target, not a clinical efficacy guarantee.
- Two active arms vs. one shared placebo raises a multiplicity question with no public testing-hierarchy disclosure.



Ianalumab validates week-48 ESSDAI as a feasible endpoint, with modest effect sizes

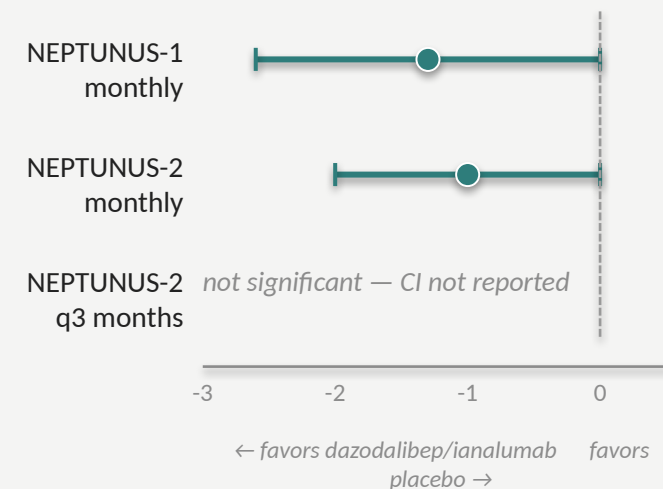
EXHIBIT — INDIVIDUAL PHASE 3 RESULTS

Trial / regimen	Effect	95% CI	p-value
NEPTUNUS-1, monthly	-1.3	-2.6 to 0.0	0.0496
NEPTUNUS-2, monthly	-1.0	-2.0 to 0.0	0.041
NEPTUNUS-2, every 3 months	Not significant (n.s.)	—	—

- Monthly treatment met the endpoint in both trials; less-frequent dosing did not.
- Ph2b met its prespecified dose-response objective (4 of 5 models significant); the 300mg pairwise comparison alone ($p \approx 0.092$) does not define trial failure.

FOREST PLOT — EFFECT VS. PLACEBO

(ESSDAI points, 95% CI)



Registry lists NEPTUNUS-2 N=506; conference presentation cites N=504 — this discrepancy is unresolved, reported both rather than picking one.

Same-pathway evidence supports systemic activity more clearly than universal symptom benefit

Evidence	Result	What it contributes
Iscalimab systemic cohort, N=173	Met dose-response objective	Supports CD40 pathway plausibility
Iscalimab symptom cohort, N=100	ESSPRI -0.57; p=0.12	Symptom benefit not assured by pathway alone
Abatacept ASAP-III	Missed continuous ESSDAI primary	Plausible biology can still fail clinically
Rituximab TRACTISS	Missed symptom-VAS responder primary	Endpoint/population selection matters
Tocilizumab ETAP	Missed composite-response primary	Distinct response construct, not continuous ESSDAI

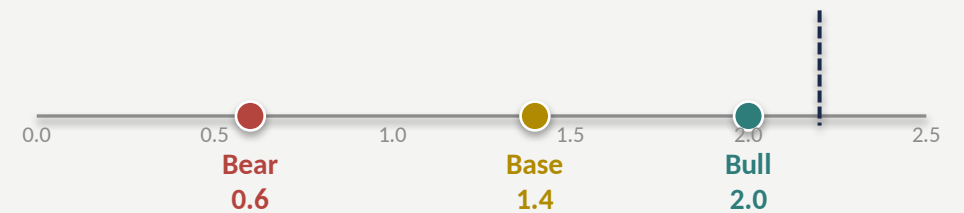
ON-SLIDE TAKEAWAY: use prior trials to challenge assumptions; avoid a numerical “class success rate.”

A 1.4-point base assumption retains about two-thirds of the Phase 2 effect

Case	Assumed benefit	Retention vs. 2.2	Rationale
Bull	2.0	91%	Most of the Phase 2 signal persists
Base	1.4	64%	Meaningful attenuation without loss of activity
Bear	0.6	27%	Substantial attenuation via efficacy, measurement, or population differences

The base case is a judgment call, not a fitted or empirically estimated mean — tested across a range rather than one decimal point.

ASSUMED BENEFIT (ESSDAI POINTS)



Three scenario points on the assumed-benefit axis, with Phase 2's observed 2.2-point effect shown as a separate reference marker (not a fourth scenario).

The base effect has favorable power under nominal testing, but the margin narrows with stricter assumptions

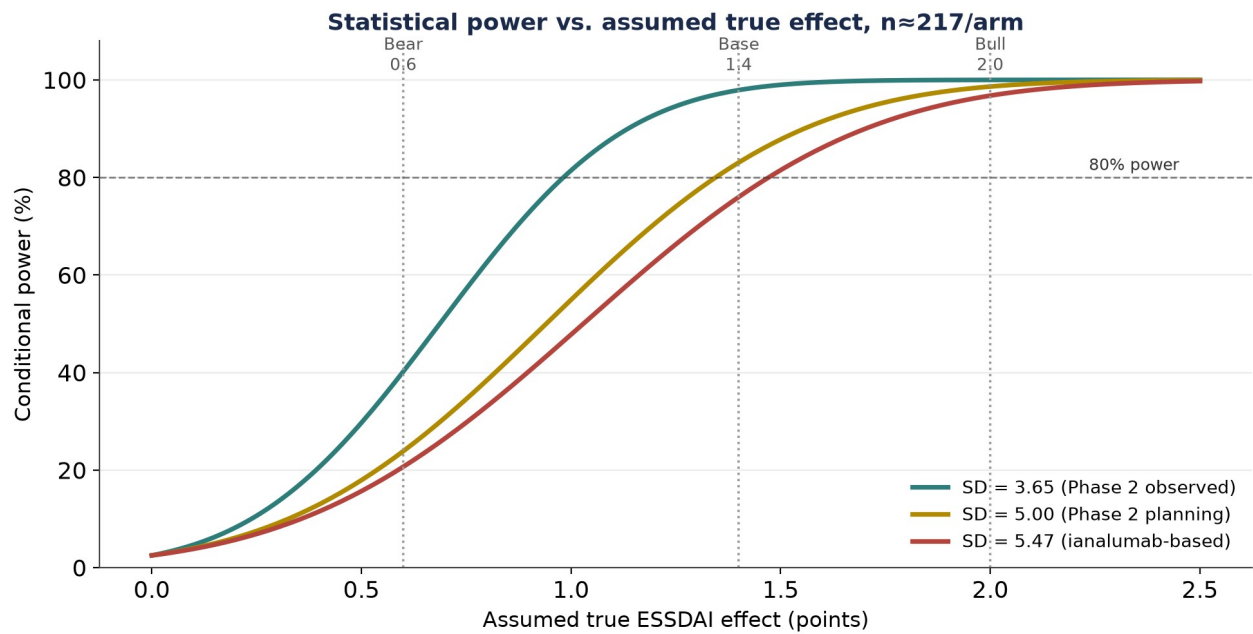


EXHIBIT — ONE ACTIVE-DOSE COMPARISON, n=217/ARM, EFFECTIVE SD=5.47

Assumed true benefit	Power @ $\alpha=0.05$	Power @ illustrative $\alpha=0.025$
0.6	21%	14%
1.0	48%	37%
1.4	76%	66%
1.6	86%	79%
2.0	97%	94%

ON-SLIDE TAKEAWAY: statistical success remains genuinely uncertain even if the assumed biological effect is correct.

Higher variance and information loss can materially weaken the same efficacy forecast

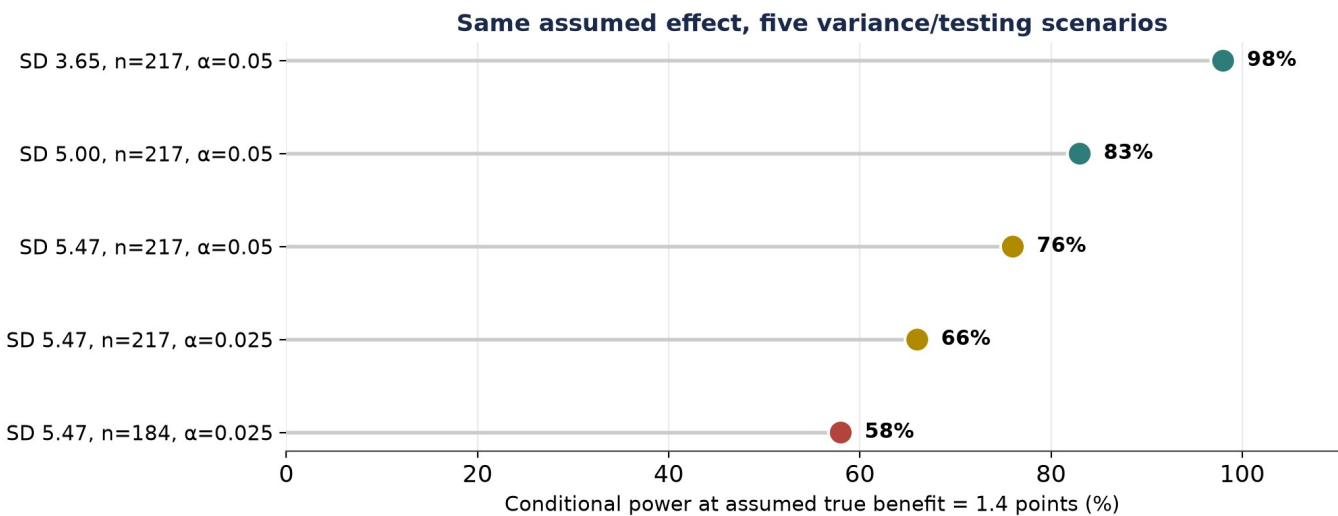


EXHIBIT — ASSUMED TRUE BENEFIT FIXED AT 1.4

Assumption	n/arm	α	Approx. power
SD 3.65	217	0.05	98%
SD 5.00 (Ph2 planning)	217	0.05	83%
SD 5.47 (ianalumab-based)	217	0.05	76%
SD 5.47, stricter α	217	0.025	66%
Same, reduced effective n	184	0.025	58%

- Raw sample size alone is insufficient — repeated measurement and adjustment affect precision.
- Phase 3’s treatment of missing data/rescue medication is unknown and matters.

ESSPRI evidence supports optimism; DASPRI is a new, FDA-PRO-aligned instrument with zero track record for this drug

Question	Evidence available	Interpretation
Does DAZ improve reported symptoms?	Positive Phase 2 ESSPRI total/domain results	Supports a positive ESSPRI lean
Does benefit extend to week 48?	Supportive crossover only, no parallel-group replication	Durability remains a test
What supports DASPRI?	Confirmed absent from the Phase 2 paper — searched full text, zero occurrences. DASPRI was developed as a patient-symptom diary specifically to align with FDA patient-reported-outcome guidance	No dazodalibep-specific precedent exists for this instrument at all; likely added for regulatory-alignment reasons, not because ESSPRI underperformed
What defines overall trial success?	Two co-primary outcomes; procedure not established	Do not assume both-pass or either-pass rule

This isn't a research gap to close — it's a real fact about the evidence base.

Fc-free engineering addresses a known mechanism; Phase 2 safety still warrants careful scrutiny

EXHIBIT — SELECTED SERIOUS OBSERVATIONS, NOT AN EXHAUSTIVE SAFETY TABLE

Observation	Population	Attribution and precise timing
DVT and drug-induced liver injury, same participant	Population 1	Both investigator-attributed as related; onset 180 days after last DAZ dose
Invasive ductal breast carcinoma	Population 2	Investigator-attributed as related; onset 139 days after last dose
One death, unknown cause following COVID-19 diagnosis	Population 1	SAE reported during Stage I per the paper’s text; investigators considered it unrelated; the death itself is tabulated under the Stage II+follow-up safety window (46 days post-last-dose) in Table 3 — both facts are accurate and reported together here, not simplified to one label

Three investigator-attributed serious events in two participants, plus a separately described unrelated death. Individual events in a small trial establish neither causation nor absence of risk.

Denominators: Stage I systemic DAZ n=36; Stage II systemic DAZ→placebo n=34; symptomatic safety set n=48.

Our positive systemic forecast stands or falls on the prespecified primary result

FORECAST

Expect positive ESSDAI efficacy in NCT06104124; favorable ESSPRI evidence with unresolved full-trial interpretation for NCT06245408.

Readout	Interpretation
Prespecified primary procedure met, consistent dose results	Confirms the forecast; assess magnitude and safety separately
Primary met, modest effect or mixed supportive outcomes	Statistical win with a more qualified clinical interpretation
Prespecified primary procedure not met	Forecast was wrong — evaluate why without redefining success post hoc
Efficacy positive, concerning safety pattern	Efficacy call may be right while benefit-risk remains unfavorable

Registry export: systemic primary completion July 23, 2026; study completion August 17, 2026 (both actual). Symptomatic: primary completion est. October 22, 2026; study completion est. December 17, 2026. Completion is not disclosure — no specific announcement date is established here.

Appendix 1 — Registry facts and unresolved design fields

Field	Systemic	Symptomatic
NCT	NCT06104124	NCT06245408
Enrollment	651	434
Status	Completed	Active, not recruiting
Primary outcomes	ESSDAI	ESSPRI; DASPRI
Primary assessment	Week 48	Week 48
Primary completion	2026-07-23 (actual)	2026-10-22 (estimated)
Study completion	2026-08-17 (actual)	2026-12-17 (estimated)

Fields not represented by the registry export: arm doses/frequency, allocation ratio, full eligibility text, background-treatment stability, rescue rules, complete outcome definitions, protocol/SAP, analysis population, actual/estimated date flags beyond what's shown. Do not present this table as a complete protocol abstraction.

Appendix 2 — Baseline characteristics (verified directly against Table 1, page 4)

Characteristic	Pop 1 Placebo	Pop 1 DAZ	Pop 2 Placebo	Pop 2 DAZ
N	38	36	55	54
Baseline ESSDAI, mean (SD)	10.1 (4.1)	11.4 (4.5)	2.5 (1.6)	3.1 (1.8)
Baseline ESSPRI total	6.6 (1.8)	6.6 (1.6)	7.1 (1.1)	7.5 (1.5)
Dryness	6.9 (2.3)	7.3 (1.7)	6.8 (1.2)	7.1 (1.6)
Fatigue	6.8 (2.2)	7.0 (1.7)	6.8 (1.8)	7.1 (1.8)
Pain	6.1 (2.3)	5.5 (3.0)	6.3 (1.9)	6.6 (2.6)
OSDI	40.0 (20.9)	48.7 (25.0)	48.6 (19.2)	48.2 (23.8)
Stimulated salivary flow, mL/min	0.97 (0.85)	0.88 (0.64)	0.83 (0.83)	1.10 (1.18)
Concomitant RA/SLE	9 (23.7%)	7 (19.4%)	7 (12.7%)	4 (7.4%)

Schirmer's criterion: ≤ 5 mm/5min in at least one eye (corrected from an earlier transcription error — this direction is confirmed against the source).

Provenance: values re-verified directly against the rendered Table 1 image three separate times across this project's review history, after two external reviews proposed alternate Pop. 2 ESSPRI-total/Dryness/OSDI/salivary-flow values. Each re-check matched the values above, not the proposed alternates (which appear to reflect a row-transposition error repeated across both reviews). Use the values above.

Relevant background therapy: Population 1 had 59.5% antimalarial, 41.9% glucocorticoid, 31.1% conventional DMARD use overall. Population 2 cholinergic agonist use differed: 11.1% DAZ vs. 27.3% placebo.

Appendix 3 — Supportive outcomes and domain scope

Day-169 outcome	Population	DAZ / Placebo change	p-value
ESSPRI	Systemic	-1.8 / -1.1	0.1110
FACIT-Fatigue	Systemic	+8.1 / +5.8	0.3028
FACIT-Fatigue	Symptomatic	+8.1 / +2.8	0.0095
OSDI	Systemic	-16.00 / -14.02	0.6583
OSDI	Symptomatic	-13.95 / -8.52	0.1936
Stimulated salivary flow	Systemic	+0.39 / +0.14	0.1330
Stimulated salivary flow	Symptomatic	+0.20 / -0.01	0.2170

These are unadjusted supportive/exploratory p-values. Positive FACIT change = improvement; negative OSDI/ESSPRI change = improvement.

Appendix 4 — Reproducible statistical method

delta = assumed true positive treatment benefit; *n* = per-arm sample size; *sigma* = assumed effective SD.

```
SE = sigma * sqrt(2 / n)
critical_z = inverse_normal_cdf(1 - alpha/2)
power(delta) = normal_cdf(delta/SE - critical_z)
effect_for_80_percent_power ≈ (critical_z + inverse_normal_cdf(0.80)) * SE
```

Sigma	n	Alpha	SE	Observed threshold	Effect for ~80% power
3.65	217	0.05	0.350	0.687	0.982
5.00	217	0.05	0.480	0.941	1.345
5.47	217	0.05	0.525	1.029	1.471
5.47	217	0.025	0.525	1.177	1.619
5.47	184	0.025	0.570	1.278	1.758

Input provenance: 3.65 from dazodalibep's own Phase 2 SEs (approximate back-calculation); 5.00 is the Phase 2 protocol's own planning SD; 5.47 is the effective SD implied by NEPTUNUS-1's reported SE. None is the actual, undisclosed dazodalibep Phase 3 variance.

*Interpretive rule: an assumed true delta produces conditional power, not a guaranteed p-value. NEPTUNUS-1's SE scaled only for sample size ($0.66 \times \sqrt{(137/217)} = 0.524$) implies that an observed 1.3-point difference would carry $p \approx 0.013$ at dazodalibep's assumed *N* — a precision-scaling illustration, not a forecast.*

Appendix 5 — Expanded comparator and safety context

This is a pointer slide: the underlying material is too extensive to reproduce legibly here. Each item below is pulled directly from its source file for the full detail.

Ianalumab — full arm-by-arm data

Ph2b dose-response model plus individual pairwise arms, and both NEPTUNUS trials, including the every-3-months miss.

Source: 02_research/ianalumab_competitor_analysis.md

Historical evidence — complete table

TRACTISS, ETAP, and ASAP-III, with corrected endpoint classifications.

Source: 02_research/historical_trial_evidence.md

Phase 2 safety — full AE table

All adverse events, not only the headline SAEs discussed on Slide 19.

Source: 02_research/phase2_results.md

Appendix 6 — Source library

- **St. Clair EW et al.** CD40 ligand antagonist dazodalibep in Sjögren’s disease: a randomized, double-blinded, placebo-controlled, phase 2 trial. *Nature Medicine* 30, 1583–1592 (2024). DOI: 10.1038/s41591-024-03009-3. Open access (CC-BY 4.0). Local: 00_reference/phase2_paper.pdf.
- **ClinicalTrials.gov, NCT06104124** — systemic Phase 3. Local: 01_data/NCT06104124.csv.
- **ClinicalTrials.gov, NCT06245408** — symptomatic Phase 3. Local: 01_data/NCT06245408.csv.
- **Bowman SJ et al.** Safety and efficacy of subcutaneous ianalumab in primary Sjögren’s syndrome: phase 2b dose-finding trial. *Lancet* 399, 161–171 (2022; online 2021). PMID 34861168.
- **Fisher BA et al.** Safety and efficacy of subcutaneous iscalimab in two distinct populations of patients with Sjögren’s disease (TWINSS). *Lancet* (2024). PMID 39096929.
- **Novartis Immunology Pipeline Event, Oct 2025** — NEPTUNUS individual-arm efficacy data.
- **Seror R et al.** ESSDAI/ESSPRI MCID and disease-activity states, *Ann Rheum Dis*.
- **Der K et al.** Population PK/PD modeling of dazodalibep. *ACR Convergence 2023 abstract*.
- **Felten et al., ETAP.** Interleukin-6 receptor inhibition in primary Sjögren syndrome. PMID 33208345. Note: verify the “p=0.14” figure against this primary source before final use — flagged as unconfirmed, see 02_research/historical_trial_evidence.md.
- **van Nimwegen et al., ASAP-III.** Primary randomized trial, *Lancet Rheumatology*.
- **TRACTISS** primary trial report.